

■ 2.1.9 Advanced Topic: Expressions as Trees

The expression $x^3 + (1 + x)^2$ consists of several nested pieces. Written out in full form, the expression is `Plus[Power[x, 3], Power[Plus[1, x], 2]]`. You can think of nested expressions like this as having a “tree” structure. (If you are familiar with “parse trees”, you will notice the analogy with the structure of expressions.)

`TreeForm` prints out expressions to show their “tree” structure.

```
In[1]:= TreeForm[x^3 + (1 + x)^2]
Out[1]//TreeForm= Plus[|
                    Power[x, 3], |
                    Power[|
                        Plus[1, x], 2]|]
```

The top node in the tree consists of a `Plus`. From this node come two “branches”, x^3 and $(1 + x)^2$. From the x^3 , there are then two other branches, x and 3 , which can be viewed as “leaves” of the tree.

All *Mathematica* expressions can be viewed as trees, although the trees are often quite complicated.

This matrix is a simple tree with just two levels.

```
In[2]:= TreeForm[{{a, b}, {c, d}}]
Out[2]//TreeForm= List[|
                    List[a, b], |
                    List[c, d]|]
```

Here is a more complicated expression.

```
In[3]:= {{a b, c d^2}, {x^3 y^4}}
Out[3]= {{a b, c d^2}, {x^3 y^4}}
```

The tree for this expression has several levels. The representation of the tree here was too long to fit on a single line, so it had to be broken in two.

```
In[4]:= TreeForm[%]
Out[4]//TreeForm=
List[|
  List[|
    Times[a, b], |
    Times[c, |
      Power[d, 2]|],
  |
  List[|
    Times[|
      Power[x, 3], |
      Power[y, 4]|],
  |]
```